R91/A_utility_ce | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 26 | update 37440 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0728	66.2037	11.416850	11.427978	11.371692	-0.1916	+0.3961	0.4060
TEST ALL	71.2938	66.1802	11.373670	11.385299	11.329932	-0.1966	+0.3866	0.3951

Dataset	n	NSS Top1 (%)	R91 Top1 (%)	NSS cost	R91 cost
TSP TEST	4500	45.888889	46.977778	7.970400719	7.970233556
CVRP TEST	4500	45.977778	47.533333	18.004802368	18.009811415
TSPLIB	49	20.408163	20.408163	8.217907074	8.213942005
CVRPLIB	100	70.000000	63.000000	67.101111717	67.245109663

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: legal-masked utility CE

R91/A_utility_ce | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0728	66.2037	11.416850	11.427978	11.371692	-0.1916	+0.3961	0.4060	per-problem TRAIN SBS
TSP	46.8444	33.1111	8.024368	8.045354	7.990533	-0.2608	+0.4234	0.4133	T2T
CVRP	46.9778	18.7111	18.159403	18.269377	18.064087	-0.6020	+0.5277	0.6991	RELD_CVRP
ATSP	76.7111	47.4889	1.639099	1.676905	1.632471	-2.2545	+0.4060	0.4031	UNICO_MatPOENet
OVRP	40.9778	35.7556	7.461709	7.469579	7.415777	-0.1054	+0.6194	0.6316	MoSES_RF
VRPB	81.0889	80.8444	9.353845	9.354876	9.318750	-0.0110	+0.3766	0.3834	reld_mt
VRPL	38.1333	33.0889	12.431186	12.437959	12.383444	-0.0545	+0.3855	0.3917	MoSES_RF
VRPTW	79.9111	79.9556	18.819255	18.819431	18.766139	-0.0009	+0.2830	0.3000	reld_mt
OVRPTW	83.8222	83.7778	10.143315	10.143730	10.116859	-0.0041	+0.2615	0.2584	reld_mt
OVRPB	88.5111	88.5111	6.357217	6.357254	6.341078	-0.0006	+0.2545	0.2644	reld_mt
OVRPL	41.1556	35.9111	7.438168	7.445709	7.390416	-0.1013	+0.6461	0.6616	MoSES_RF
VRPBL	78.7556	78.5111	9.368836	9.371523	9.328366	-0.0287	+0.4338	0.4365	reld_mt
VRPBTW	78.5333	78.4222	19.948245	19.950430	19.851084	-0.0110	+0.4895	0.4693	reld_mt
VRPLTW	78.8444	78.7556	18.764707	18.765871	18.707843	-0.0062	+0.3040	0.3196	reld_mt
OVRPBL	89.2444	89.2000	6.369768	6.370093	6.352824	-0.0051	+0.2667	0.2780	reld_mt
OVRPBTW	83.5556	83.4444	10.661834	10.661098	10.623245	+0.0069	+0.3633	0.3413	reld_mt
OVRPLTW	83.5111	83.4889	10.157424	10.157423	10.129818	+0.0000	+0.2725	0.2763	reld_mt
VRPBLTW	79.4889	79.4000	19.794846	19.797044	19.706420	-0.0111	+0.4487	0.4363	reld_mt
OVRPBLTW	83.2444	83.2889	10.610083	10.609955	10.571302	+0.0012	+0.3668	0.3438	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R91/A_utility_ce | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2938	66.1802	11.373670	11.385299	11.329932	-0.1966	+0.3866	0.3951	per-problem TRAIN SBS
TSP	46.9778	34.5333	7.970234	7.989621	7.937593	-0.2427	+0.4112	0.4045	T2T
CVRP	47.5333	17.9333	18.009811	18.124983	17.918509	-0.6354	+0.5095	0.6667	RELD_CVRP
ATSP	76.1333	47.2444	1.636015	1.674207	1.629441	-2.2812	+0.4034	0.4033	UNICO_MatPOENet
OVRP	40.7111	33.6444	7.446241	7.455750	7.400487	-0.1275	+0.6182	0.6297	MoSES_RF
VRPB	80.6667	80.4000	9.350470	9.351810	9.315298	-0.0143	+0.3776	0.3799	reld_mt
VRPL	37.7778	32.2667	12.349441	12.356138	12.300958	-0.0542	+0.3941	0.4016	MoSES_RF
VRPTW	81.1333	80.8444	18.688455	18.690545	18.637408	-0.0112	+0.2739	0.2857	reld_mt
OVRPTW	83.9556	83.9333	10.092713	10.093063	10.068135	-0.0035	+0.2441	0.2352	reld_mt
OVRPB	88.1111	88.1333	6.371041	6.370826	6.352225	+0.0034	+0.2962	0.3039	reld_mt
OVRPL	41.9333	34.6000	7.464010	7.472542	7.418033	-0.1142	+0.6198	0.6303	MoSES_RF
VRPBL	79.7111	79.4222	9.394568	9.396379	9.355967	-0.0193	+0.4126	0.4161	reld_mt
VRPBTW	78.7778	78.7556	19.664918	19.665725	19.577666	-0.0041	+0.4457	0.4413	reld_mt
VRPLTW	79.3556	79.2444	18.801696	18.803136	18.745143	-0.0077	+0.3017	0.3182	reld_mt
OVRPBL	89.2444	89.2889	6.375701	6.375741	6.359310	-0.0006	+0.2577	0.2691	reld_mt
OVRPBTW	84.8889	84.8222	10.659032	10.659591	10.624748	-0.0052	+0.3227	0.2956	reld_mt
OVRPLTW	82.9778	82.8667	10.085446	10.086112	10.059001	-0.0066	+0.2629	0.2633	reld_mt
VRPBLTW	79.8222	79.7333	19.703900	19.707068	19.613284	-0.0161	+0.4620	0.4435	reld_mt
OVRPBLTW	83.5778	83.5778	10.662361	10.662149	10.625574	+0.0020	+0.3462	0.3240	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R91/A_utility_ce | LIB and locked provenance

Current R91 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	20.4082	10.2041	8.213942	8.240435	8.142167	-0.3215	+0.8815	0.9208
CVRPLIB	63.0000	45.0000	67.245110	67.245364	66.945093	-0.0004	+0.4482	0.5446

SHA256: fc889238acd827fa141c9e24fad07d5bc1f64f58c21a47da73b0ea9b5e543e8b

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, fixed physical-node0 TSP slot3 role, original R77 score MLP/pair differences/maximin; stable ascending-global-solver-ID tie break

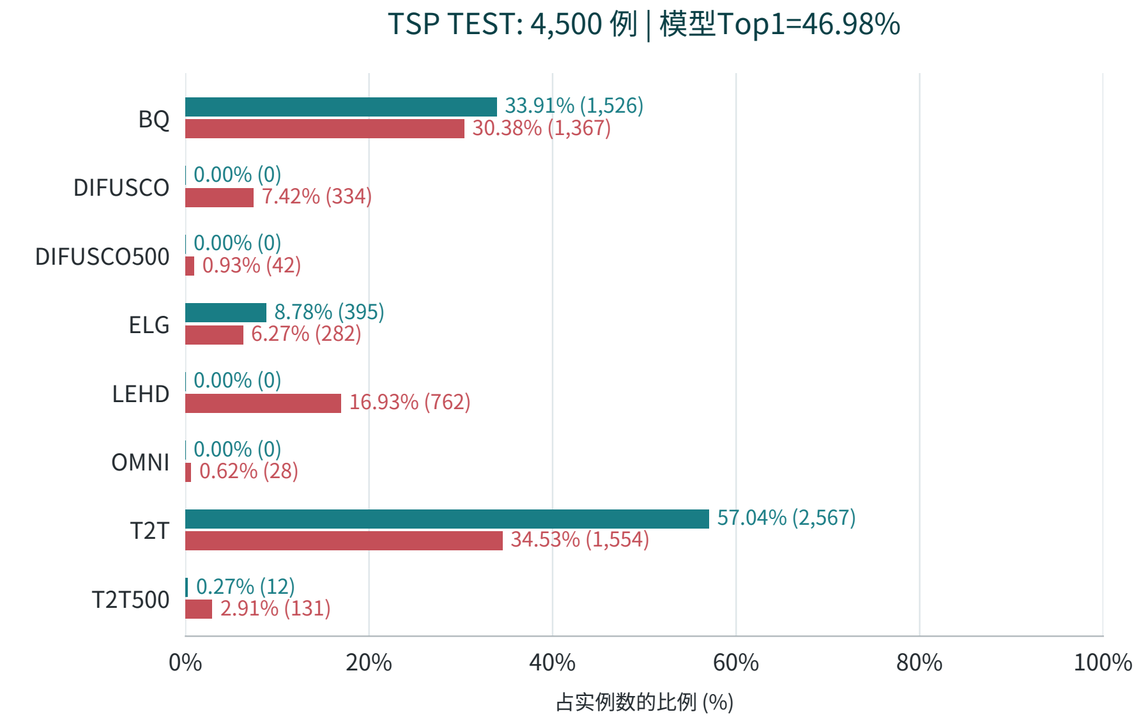
Normalization: new81000 TRAIN only; no TEST/LIB refit.

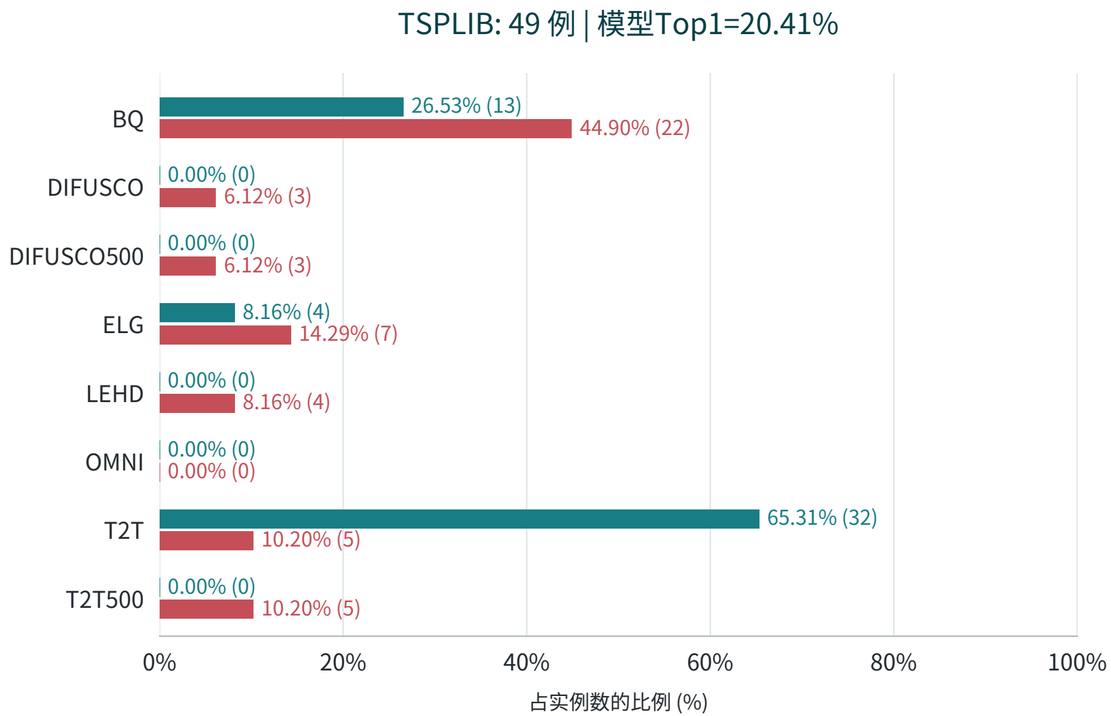
R91/A_utility_ce | TSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

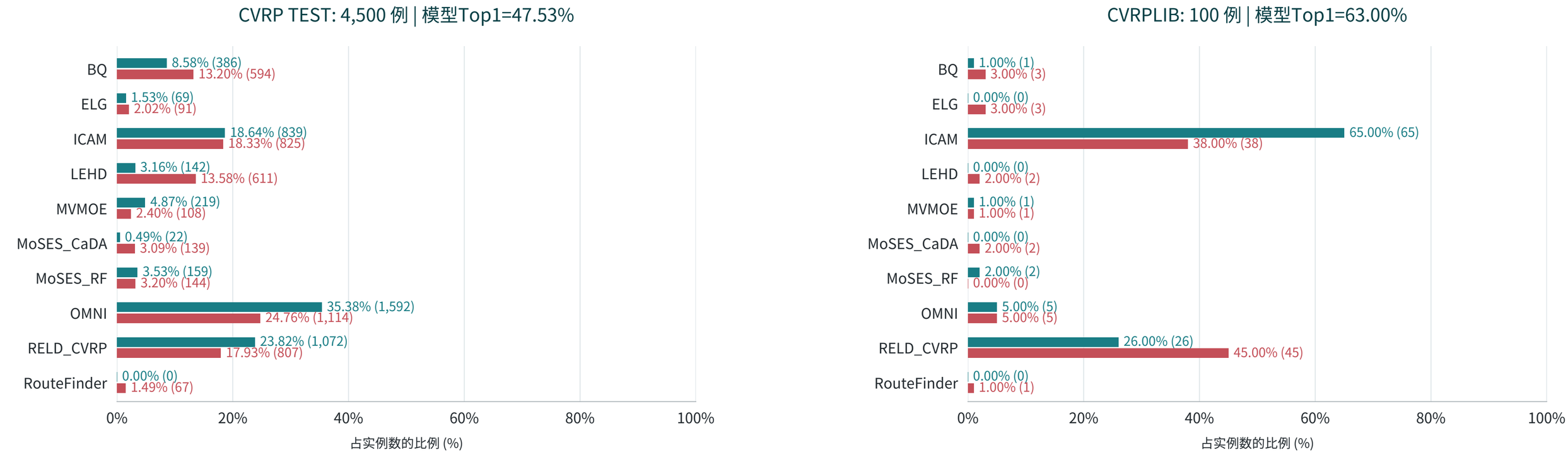
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | CVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

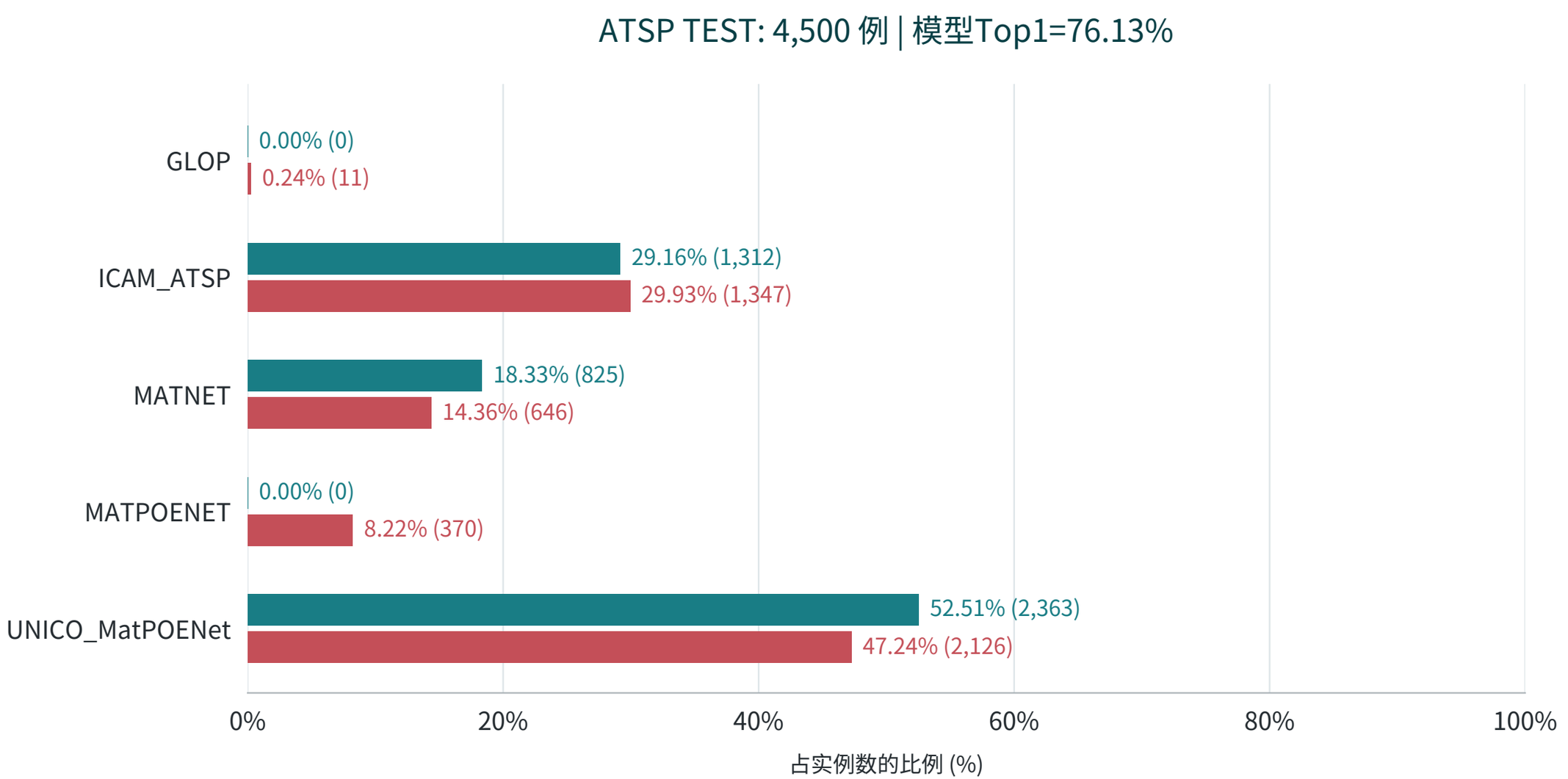
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | ATSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

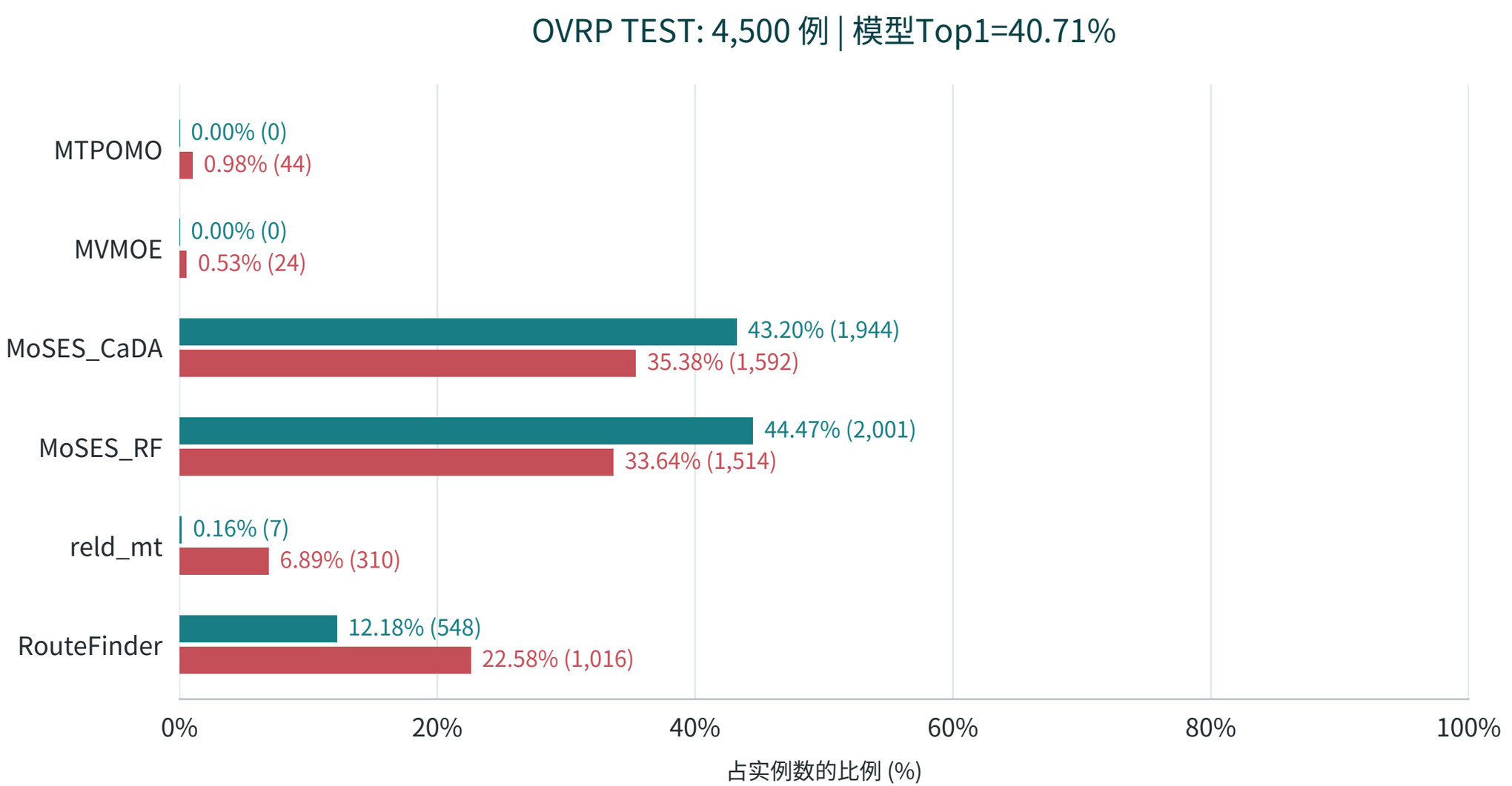
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

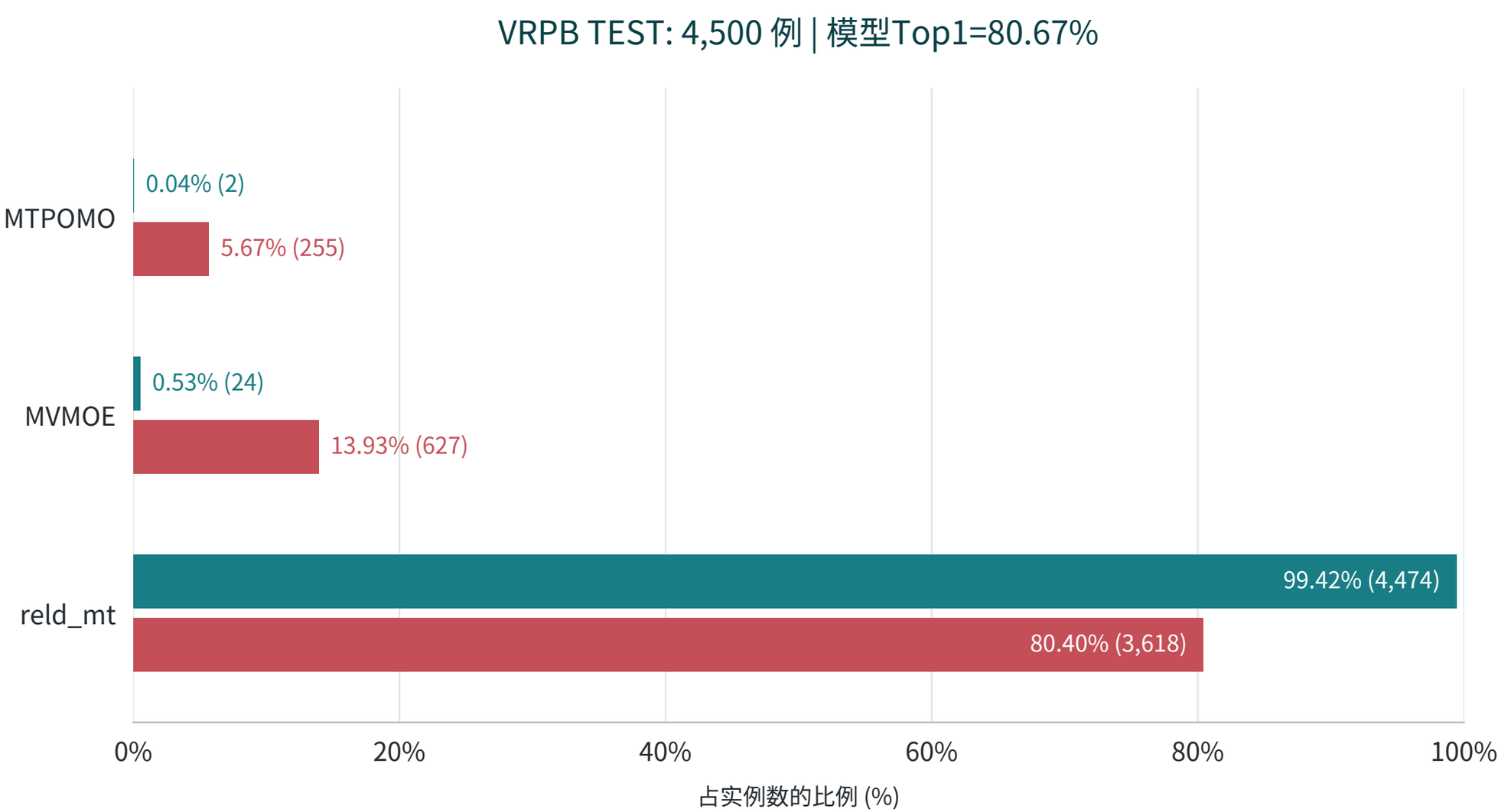
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

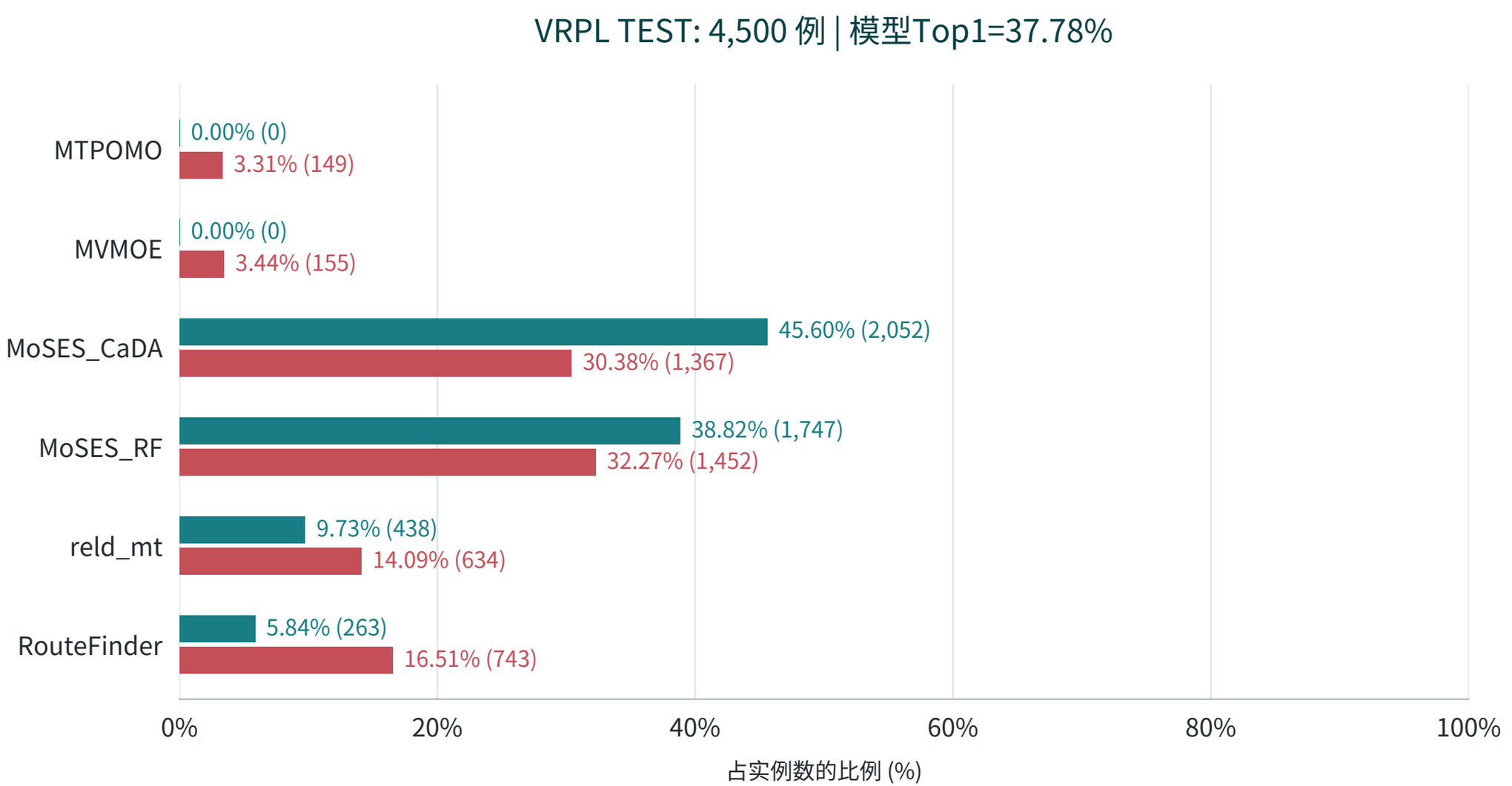
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

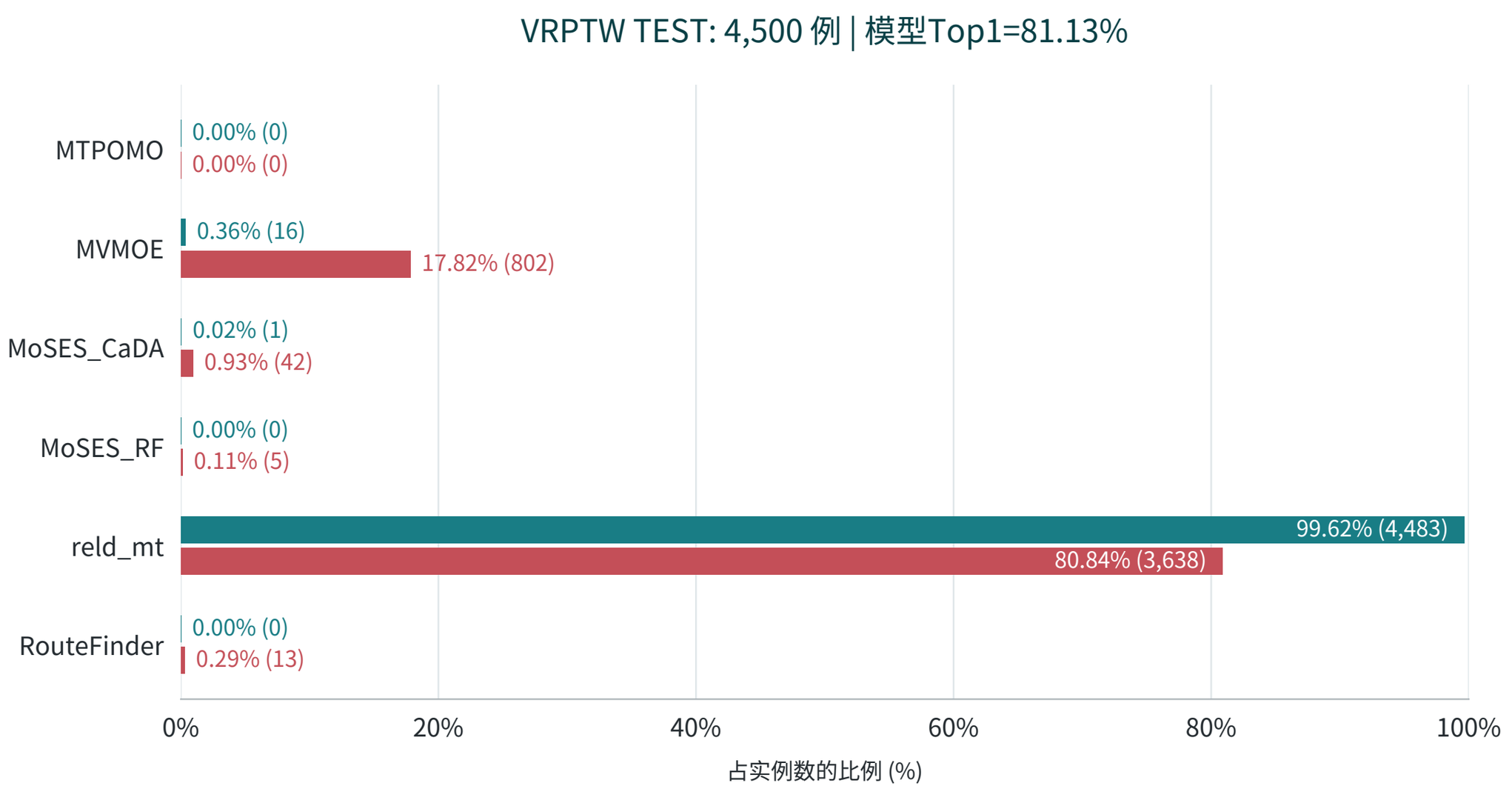
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

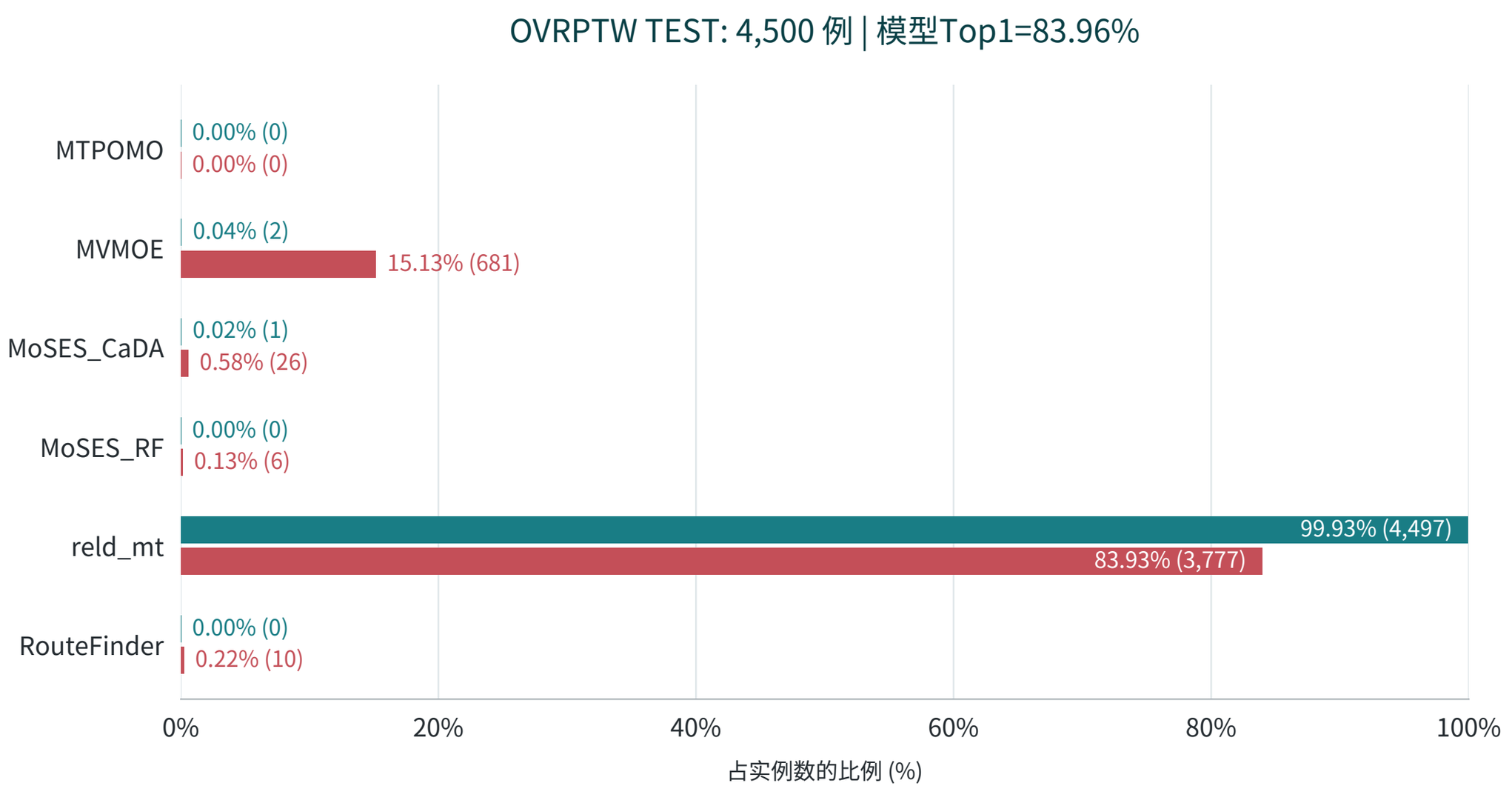
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

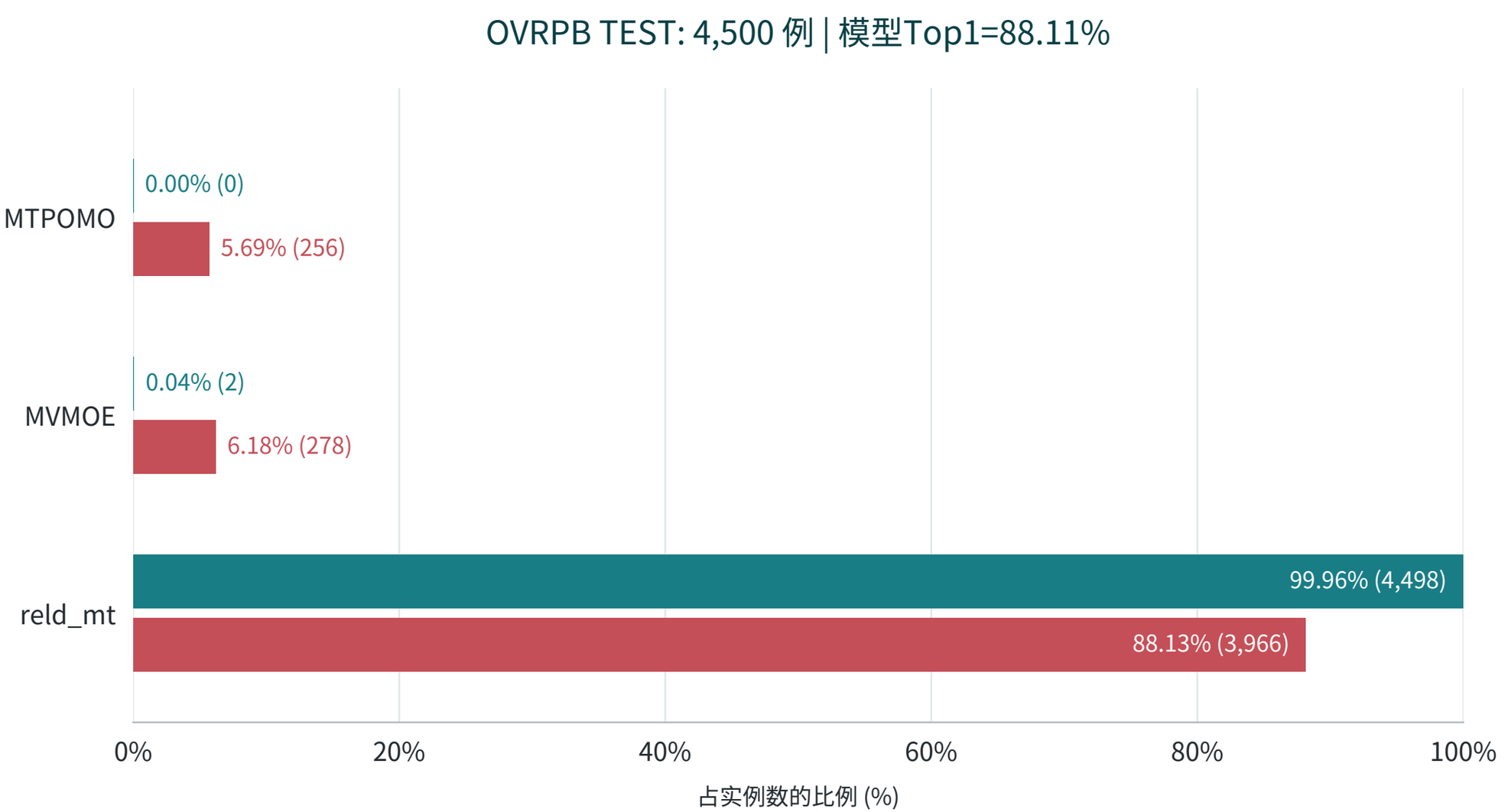
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

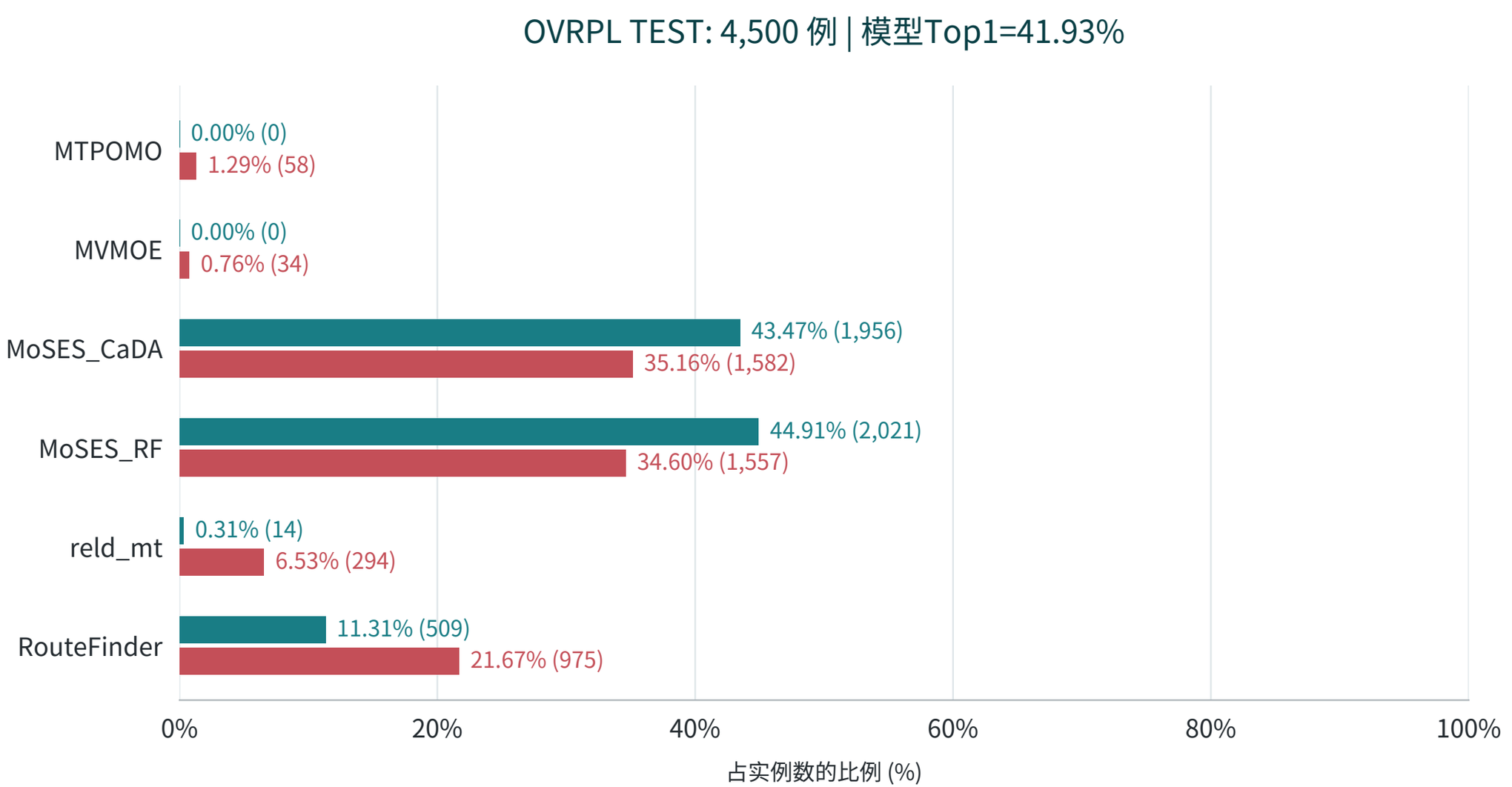
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

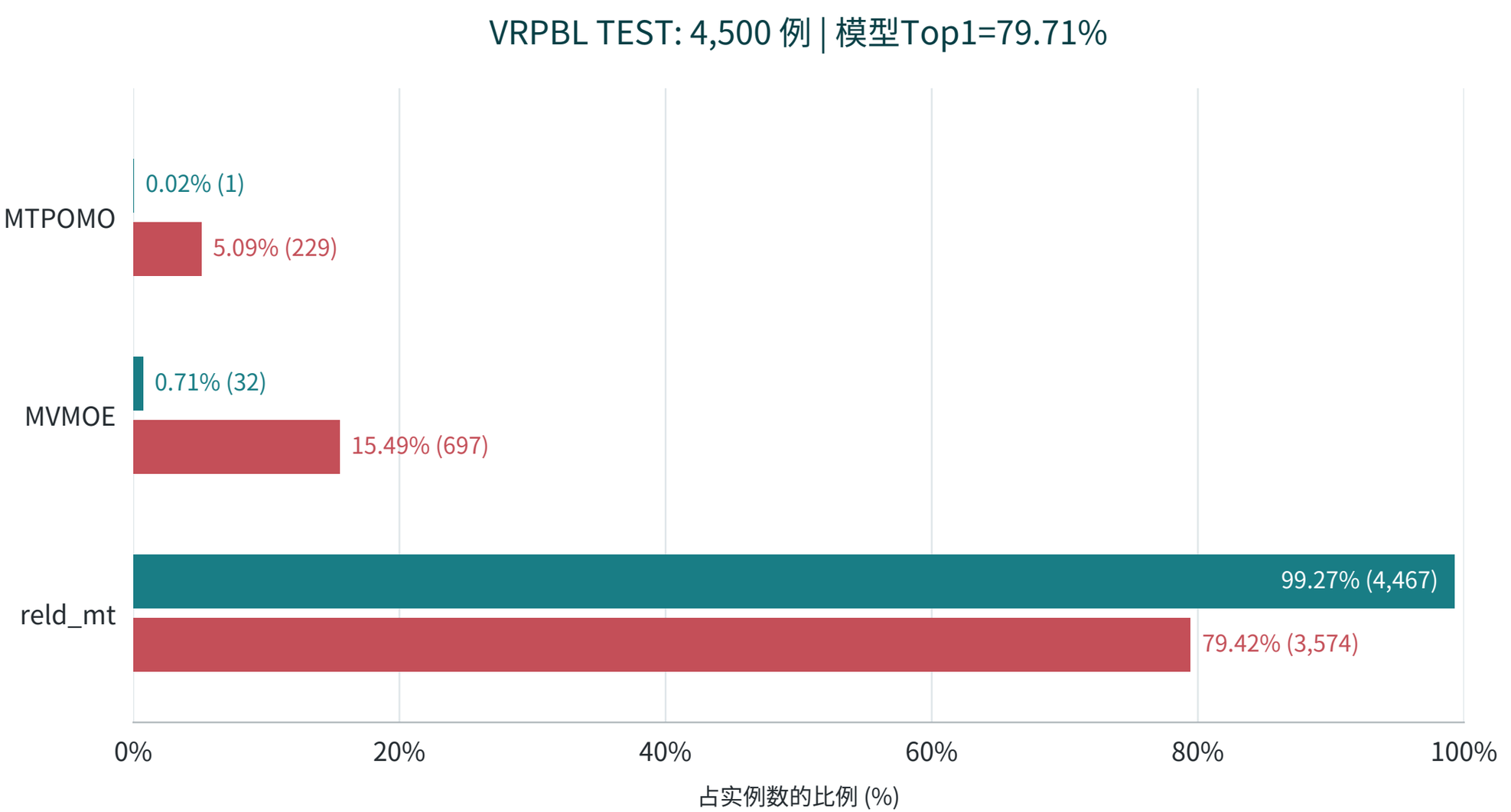
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

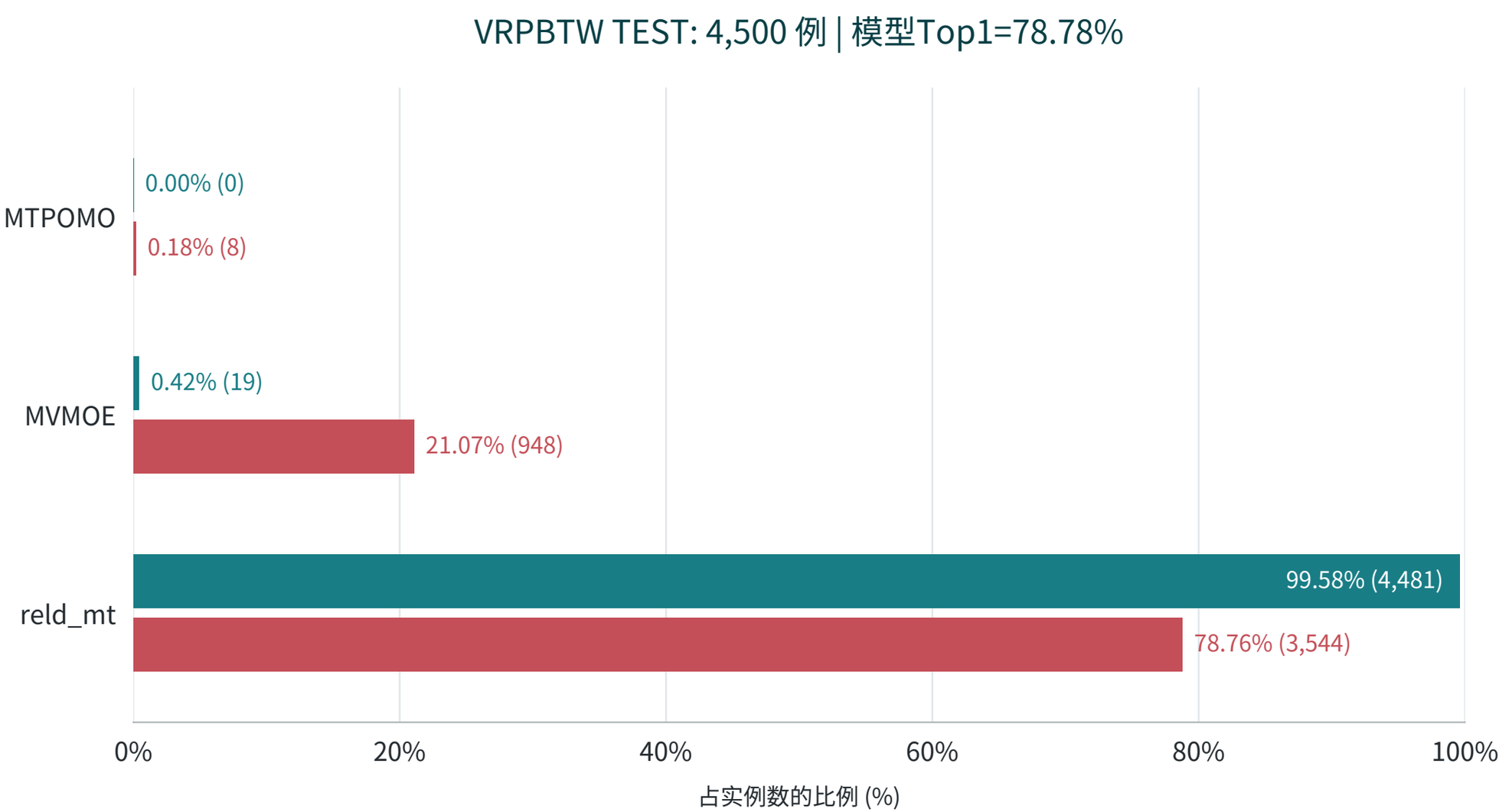
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

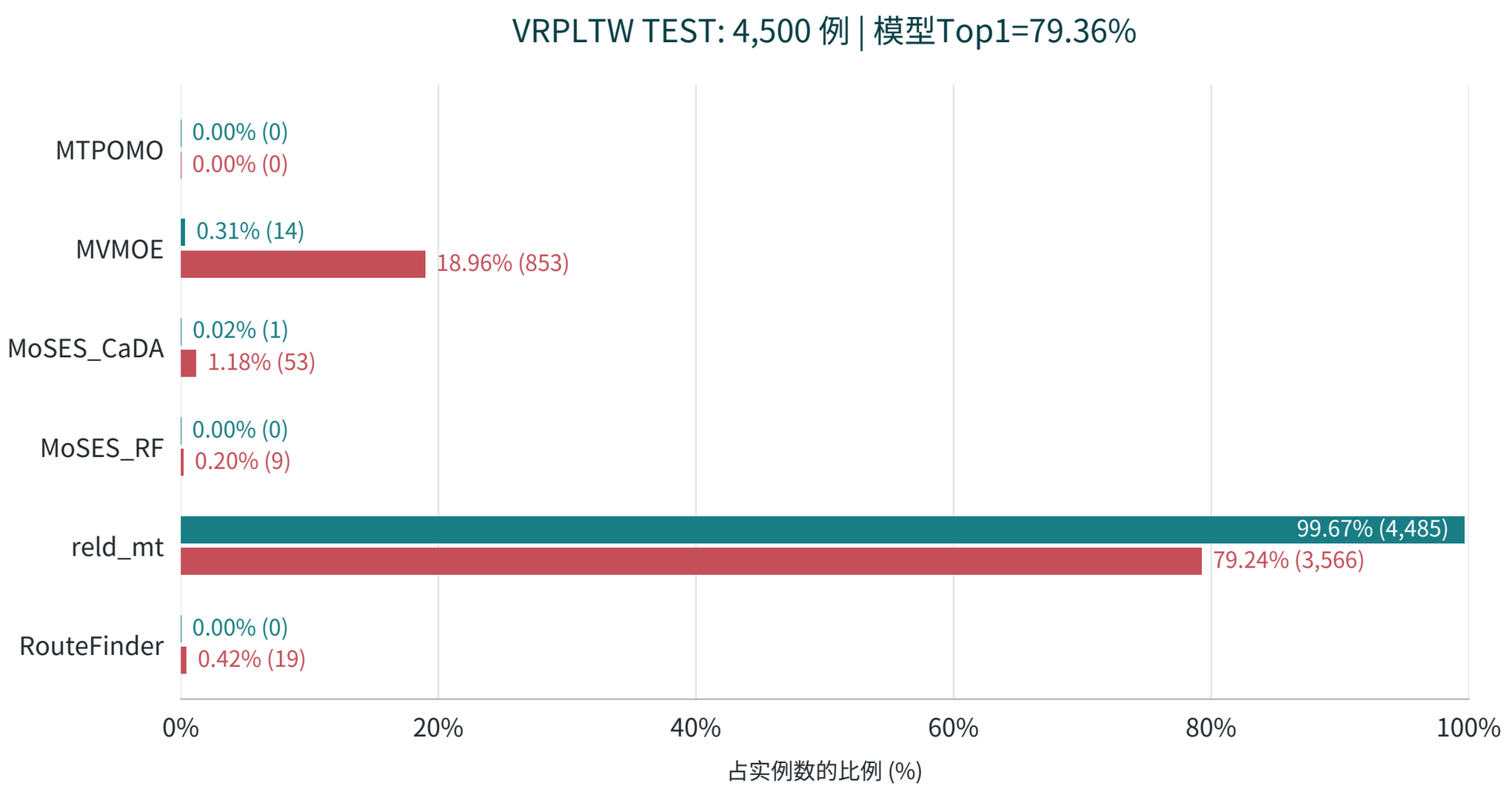
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

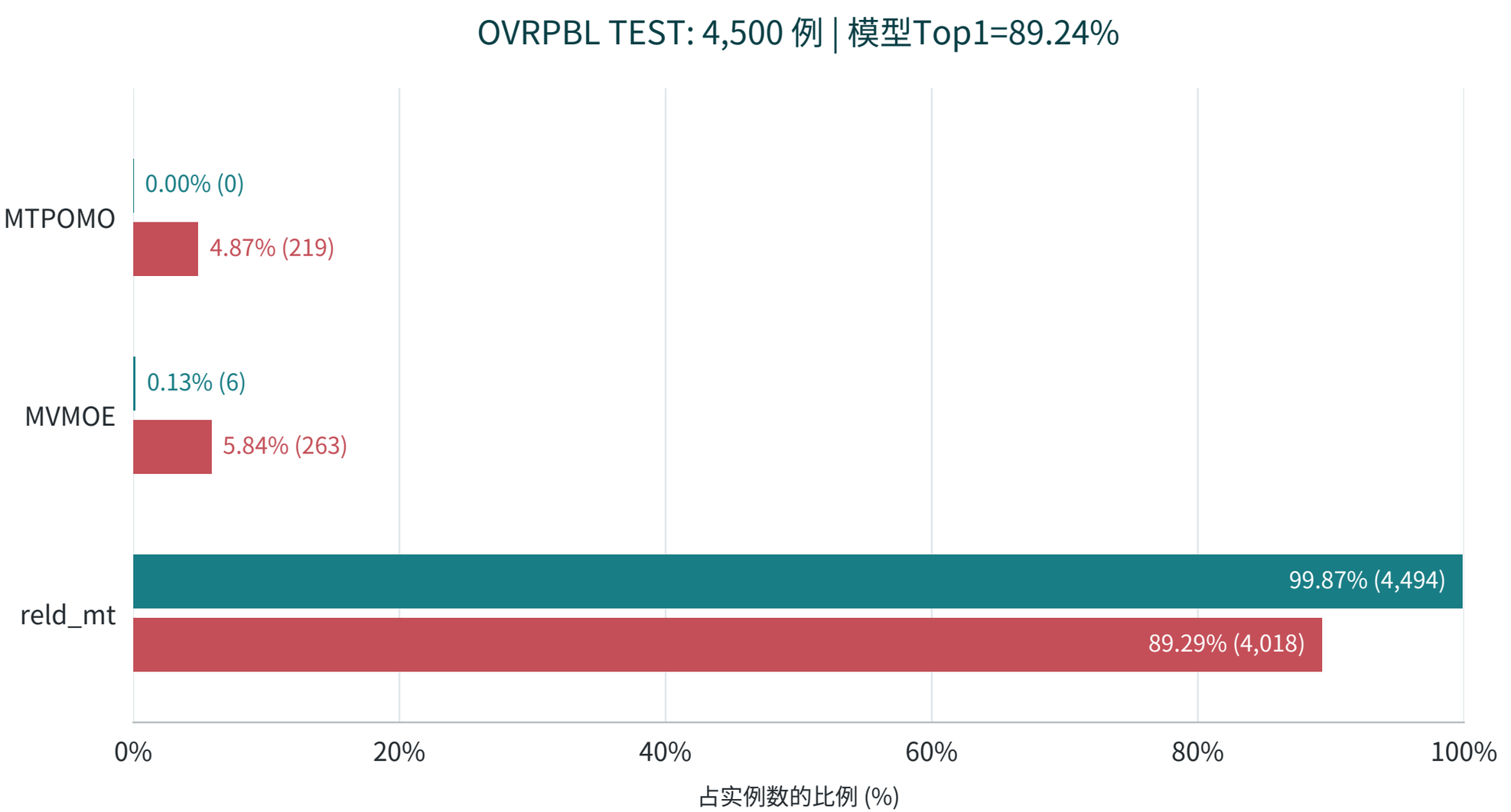
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

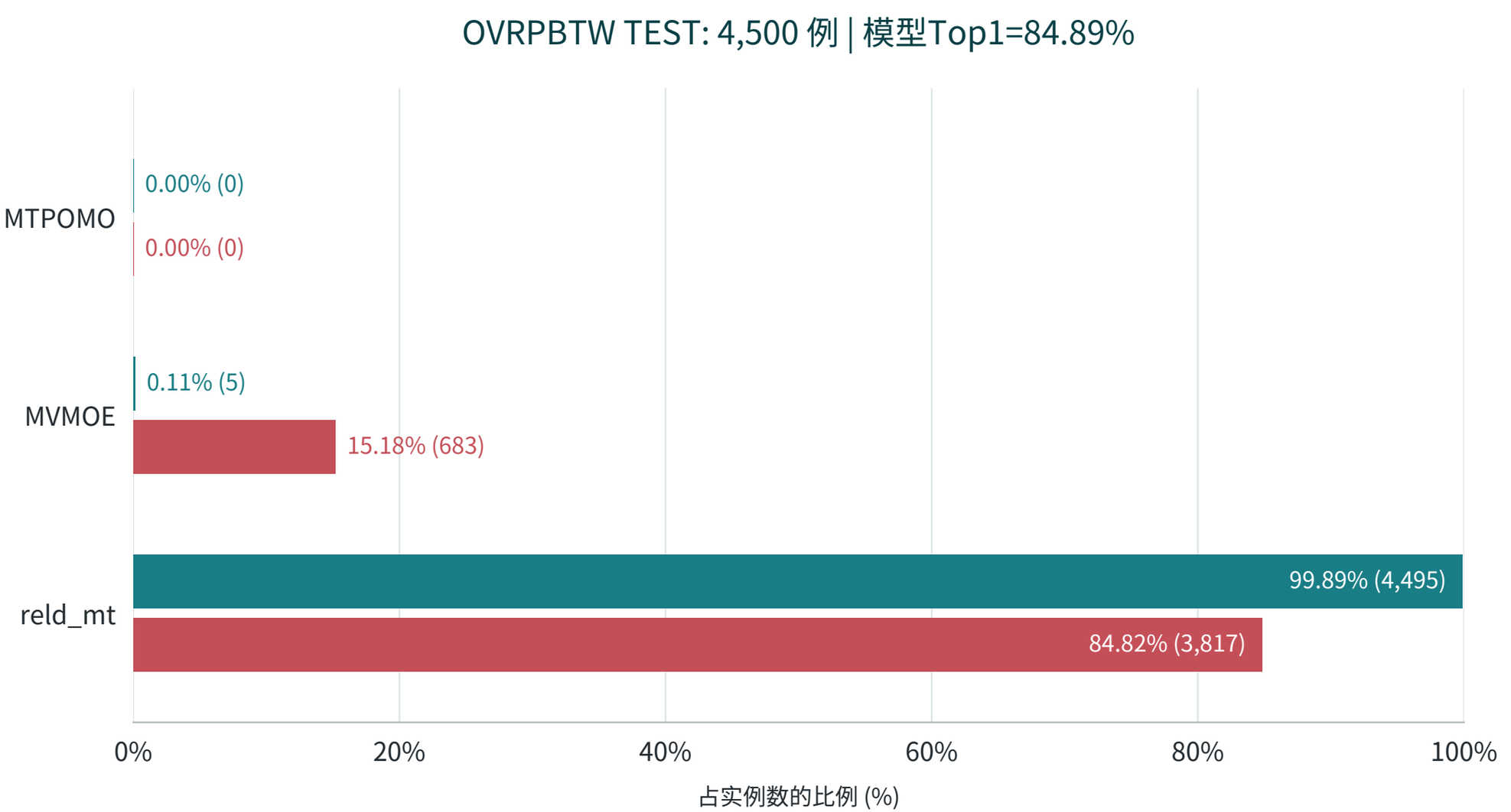
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

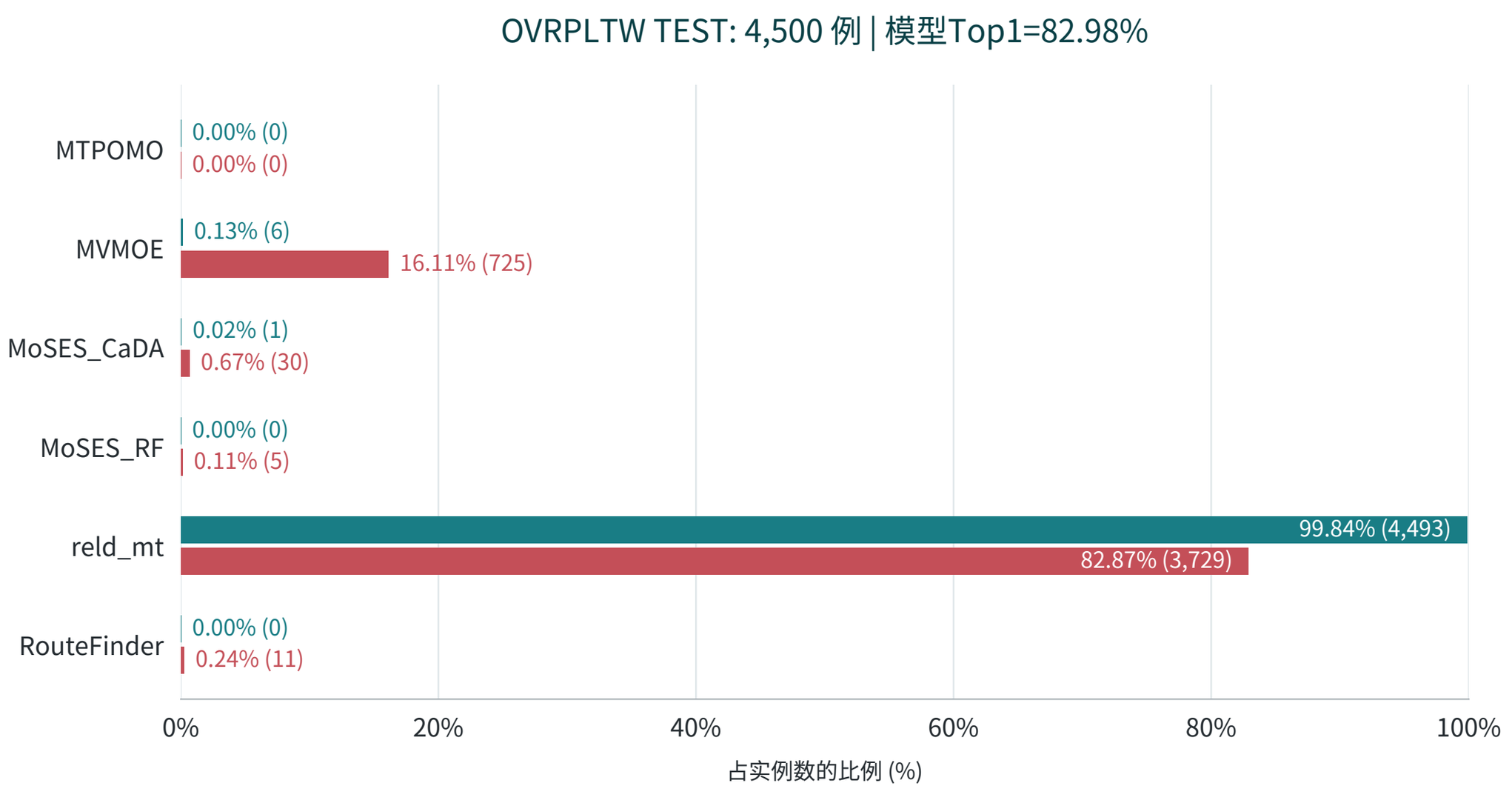
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

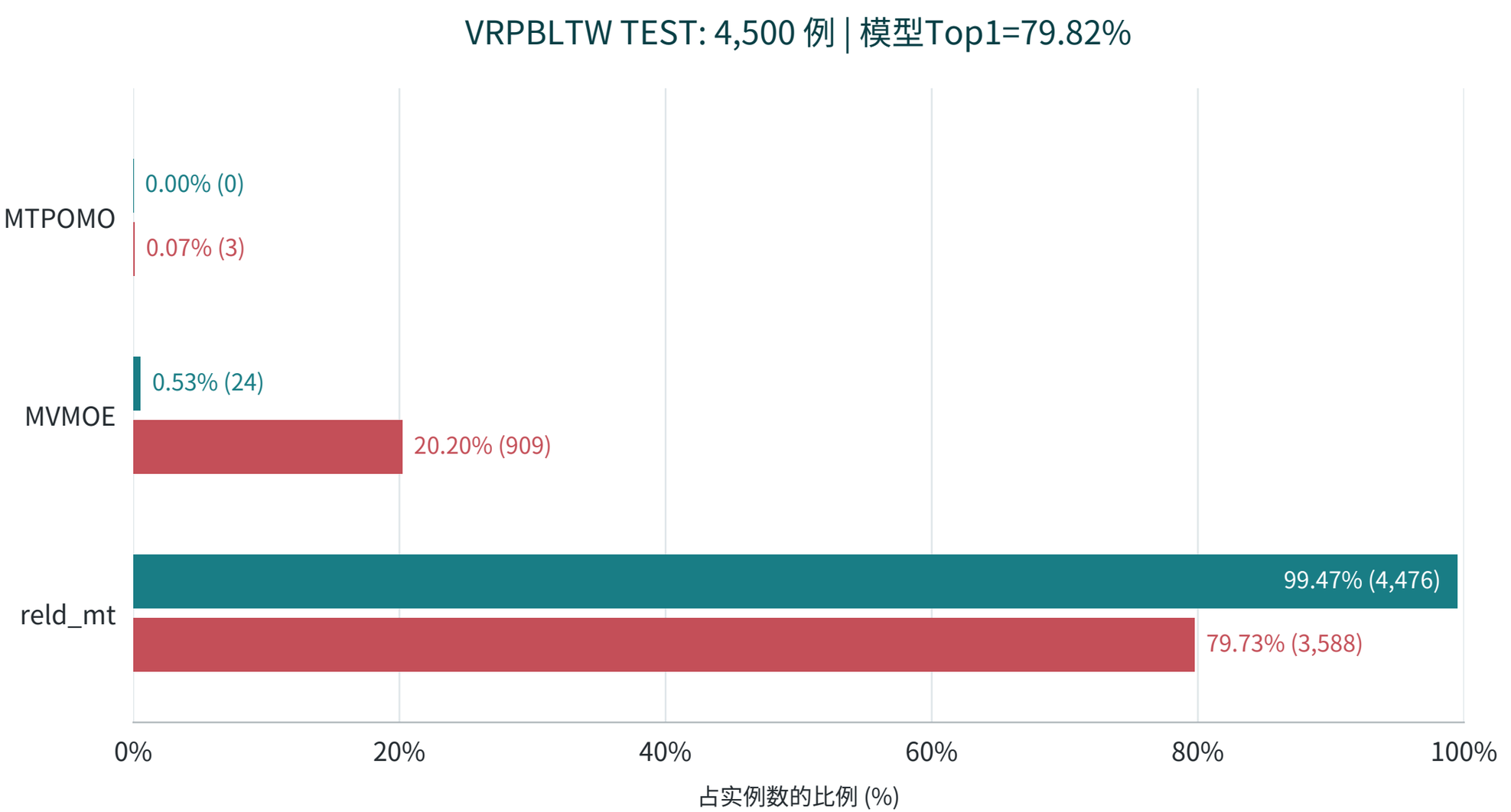
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | VRPBLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

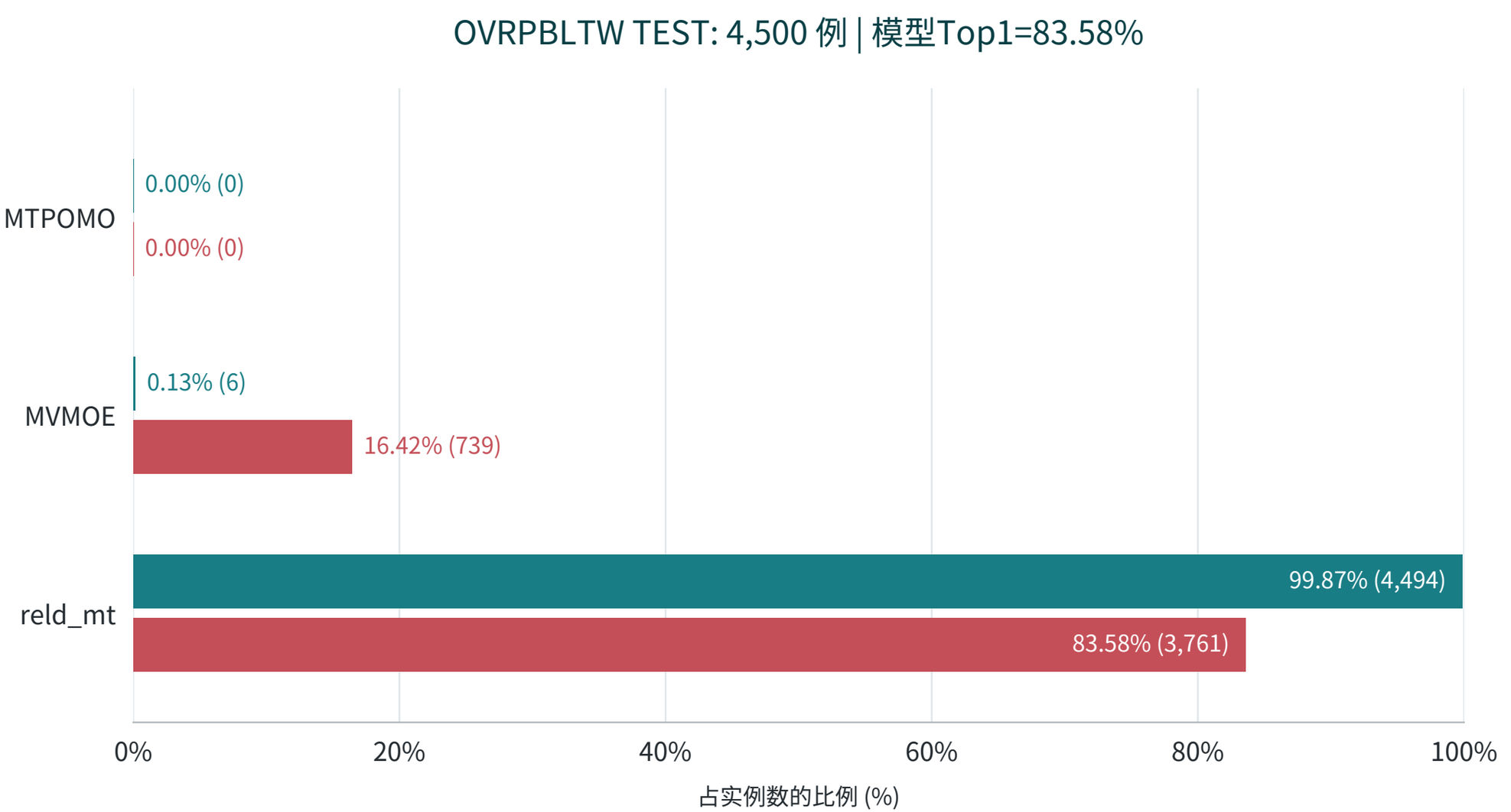
R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/A_utility_ce | OVRPBLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 fc889238acd827fa

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

R91/A_utility_ce: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.